CoolTheEngine - Configuration Summary

Engine Parameters

Chamber Pressure: 50.00 bar
Oxidizer: LOX
Oxidizer Mass Flow Rate: 2.000 kg/s
Fuel: Isopropanol
Fuel Mass Flow Rate: 1.400 kg/s

Coolant Properties

Coolant: Ethanol
Coolant Inlet Temperature: 298.0 K
Coolant Inlet Pressure: 62.00 bar
Coolant Mass Flow Rate: 1.400 kg/s

Wall Material Properties

Wall Material: CuCrZr
TBC Thickness: 0.0 µm
Channel Roughness: 30.0 µm
Wall Thickness: 1.00 mm
Total Wall Thickness: 7.00 mm

Channel Configuration

Number of Channels: 66

Channel Widths:

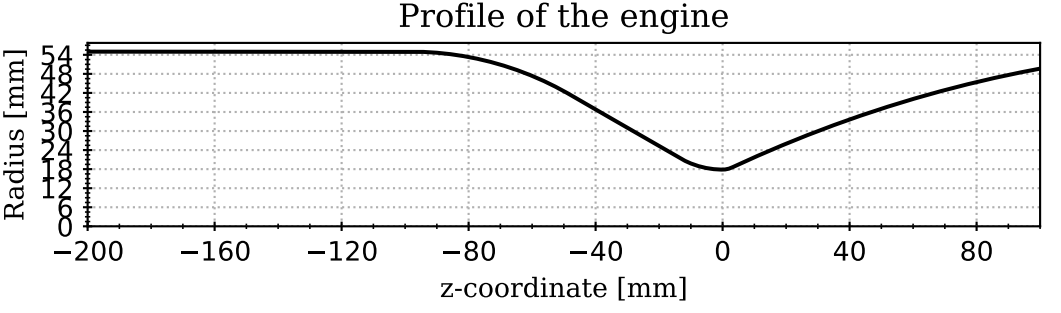
Injection: 2.50 mm
Converging: 2.50 mm
Throat: 1.00 mm
Exit: 2.00 mm

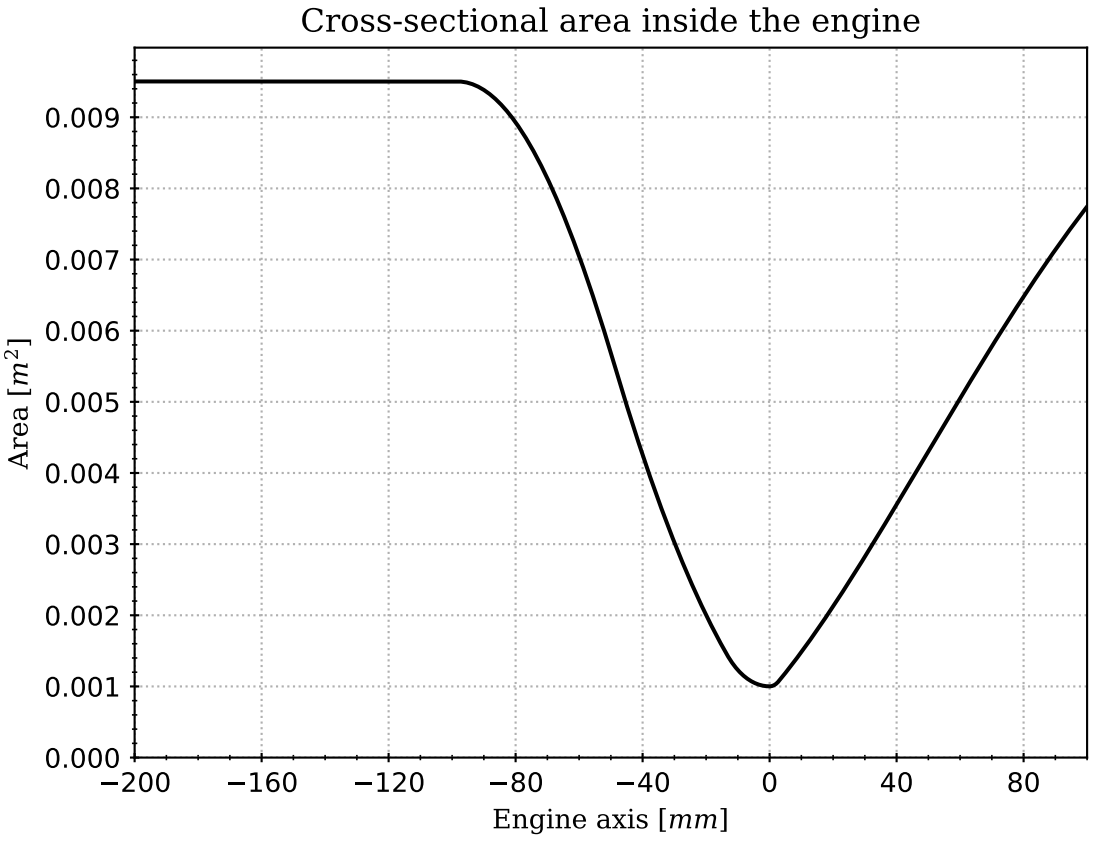
Channel Heights:

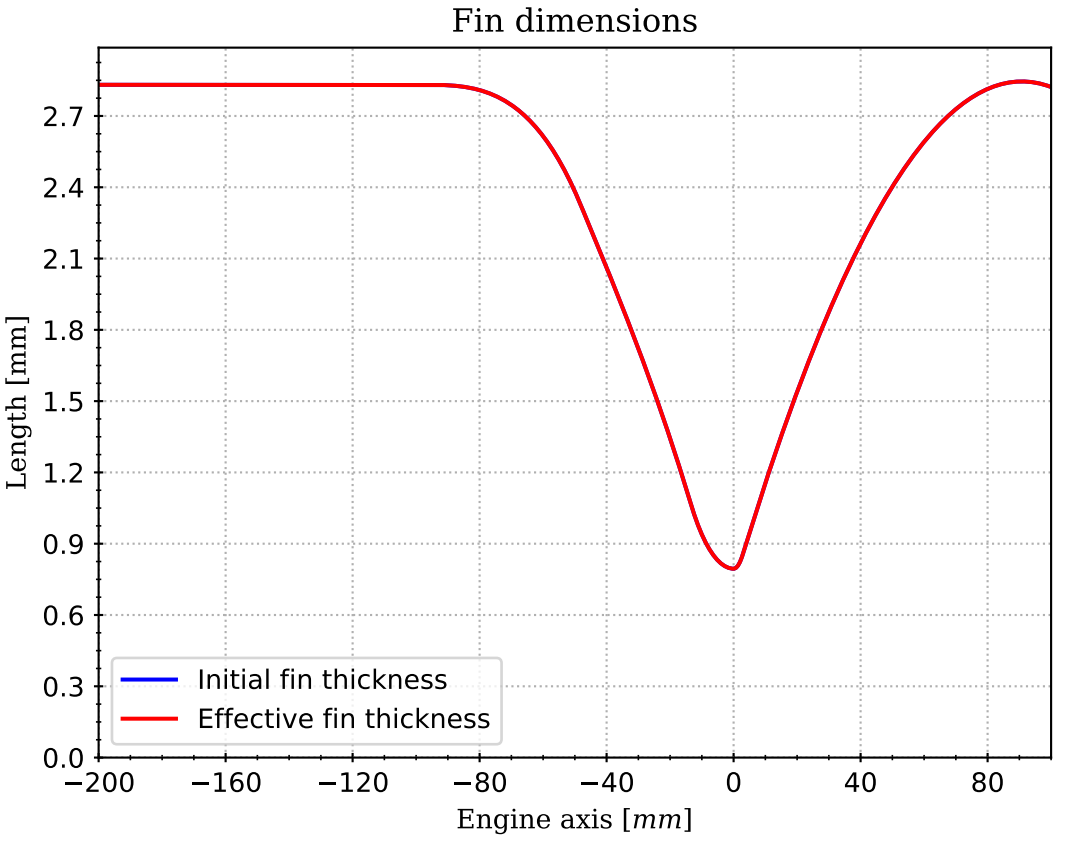
Injection: 3.00 mm
Converging: 3.00 mm
Throat: 1.00 mm
Exit: 2.00 mm

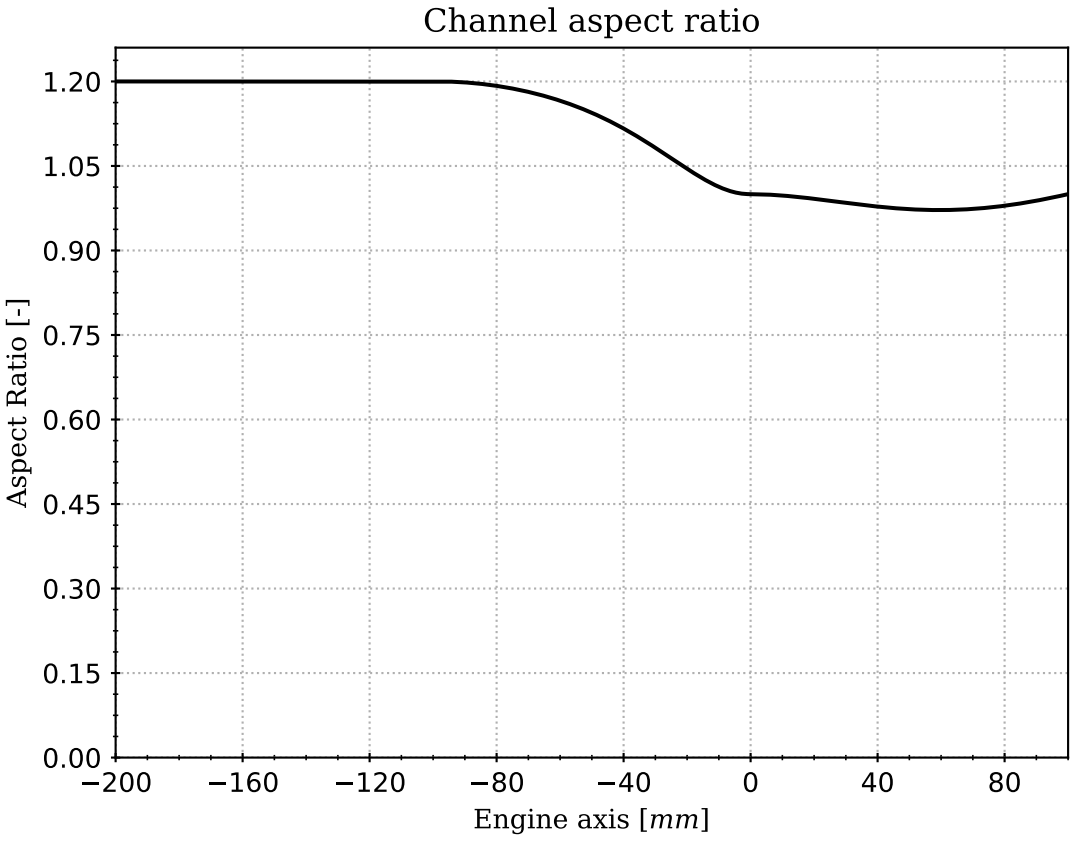
Channel Angles:

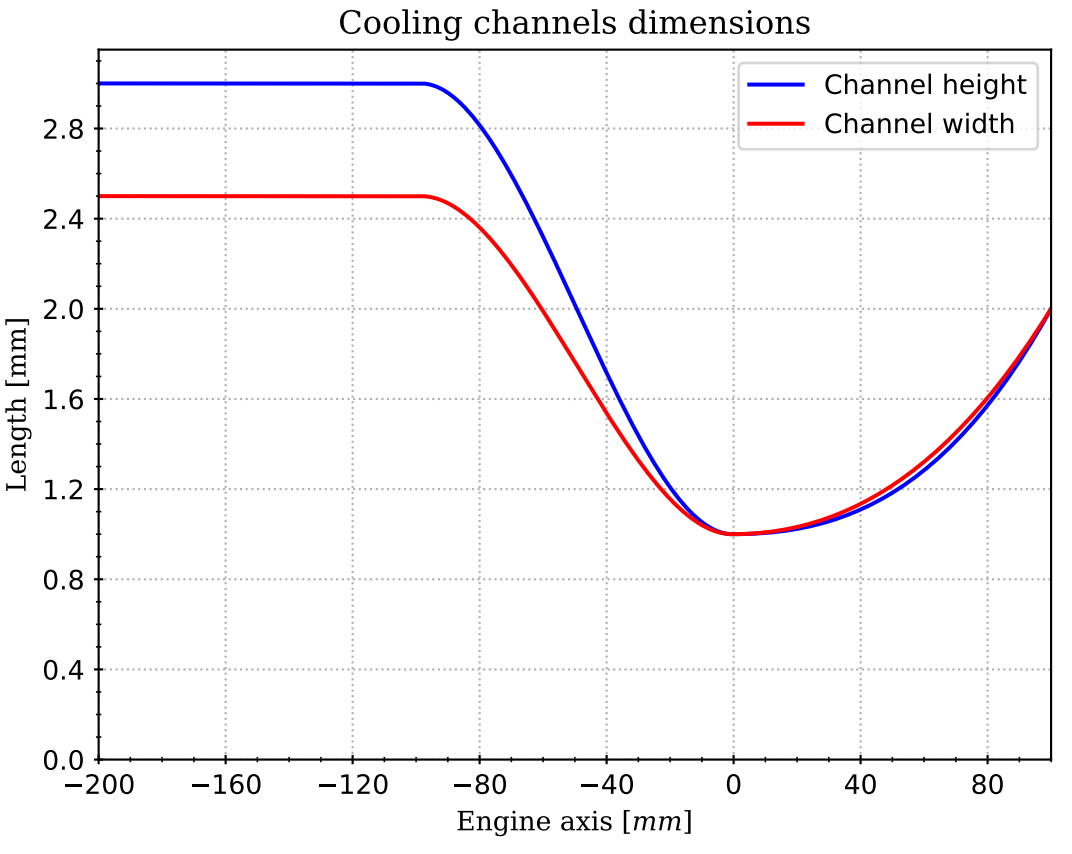
Injection: 0.0°
Converging: 0.0°
Throat: 0.0°
Exit: 0.0°

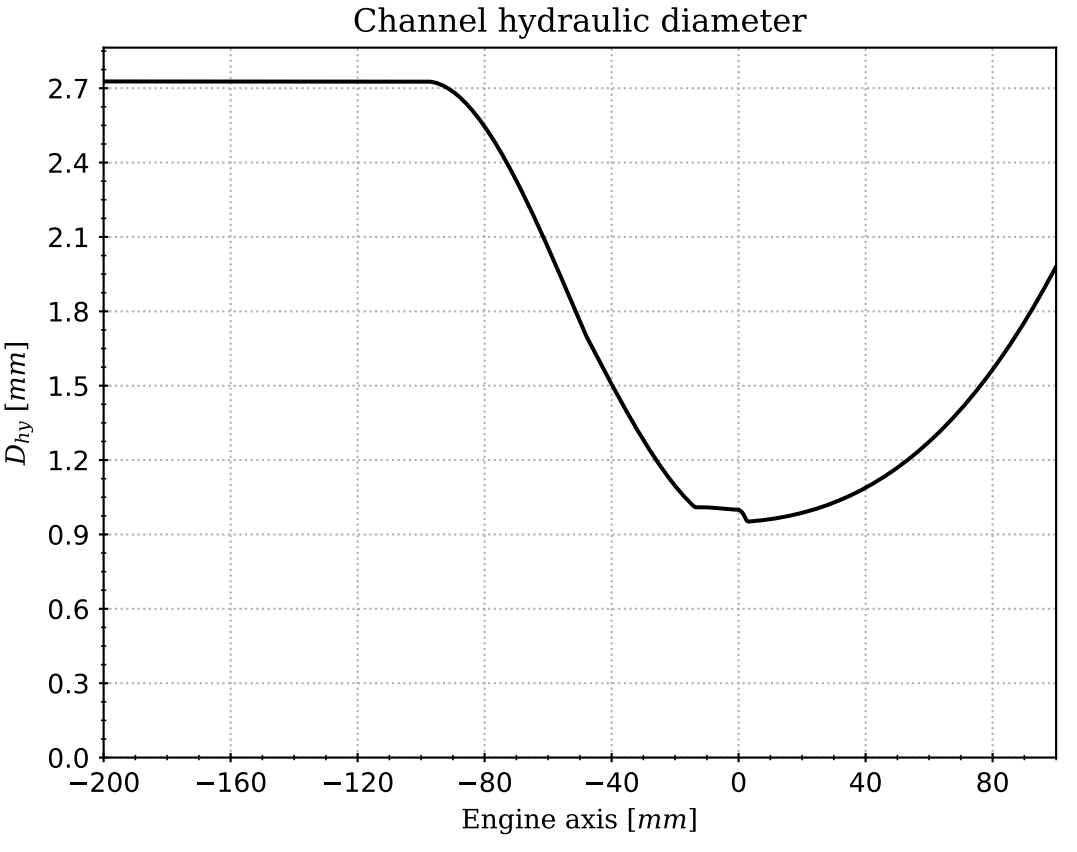

Profile of the engine
Radius [mm]
54
48
42
36
30
24
18
12
6
0
−200
−160
−120
−80
−40
0
40
80
z-coordinate [mm]

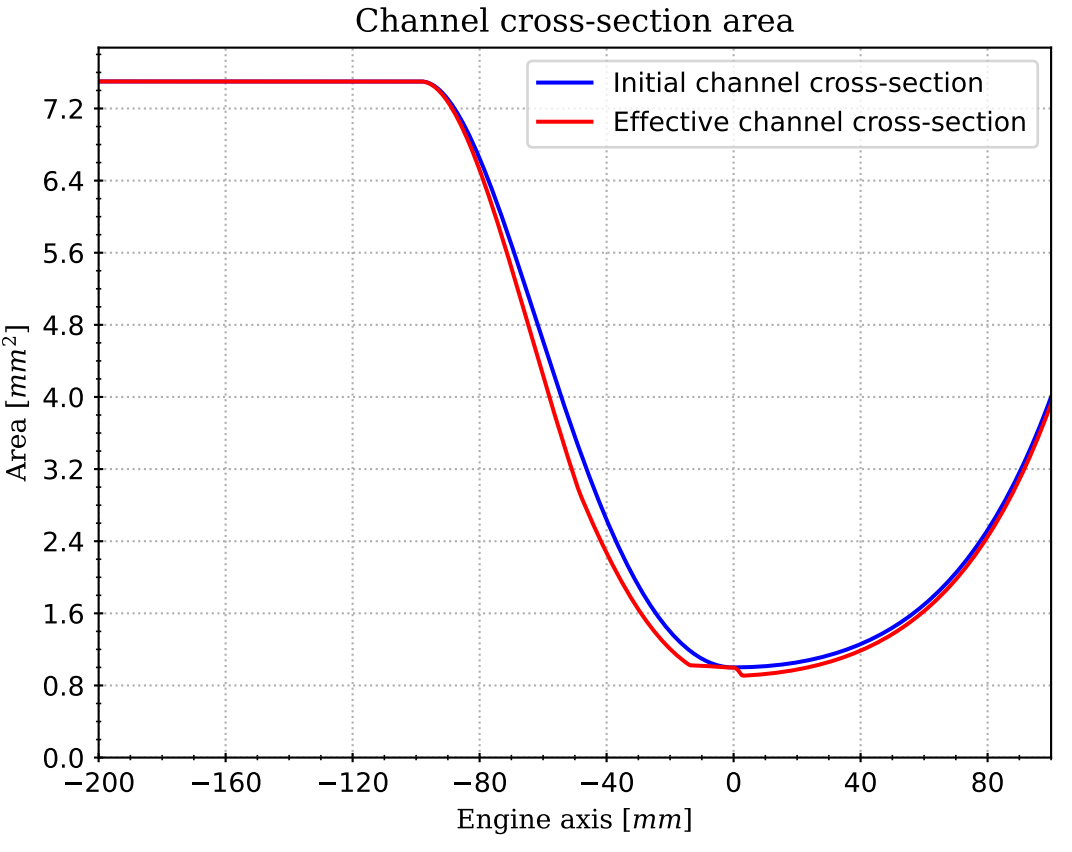

Cross-sectional area inside the engine
Area [m^2]
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
0.009
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

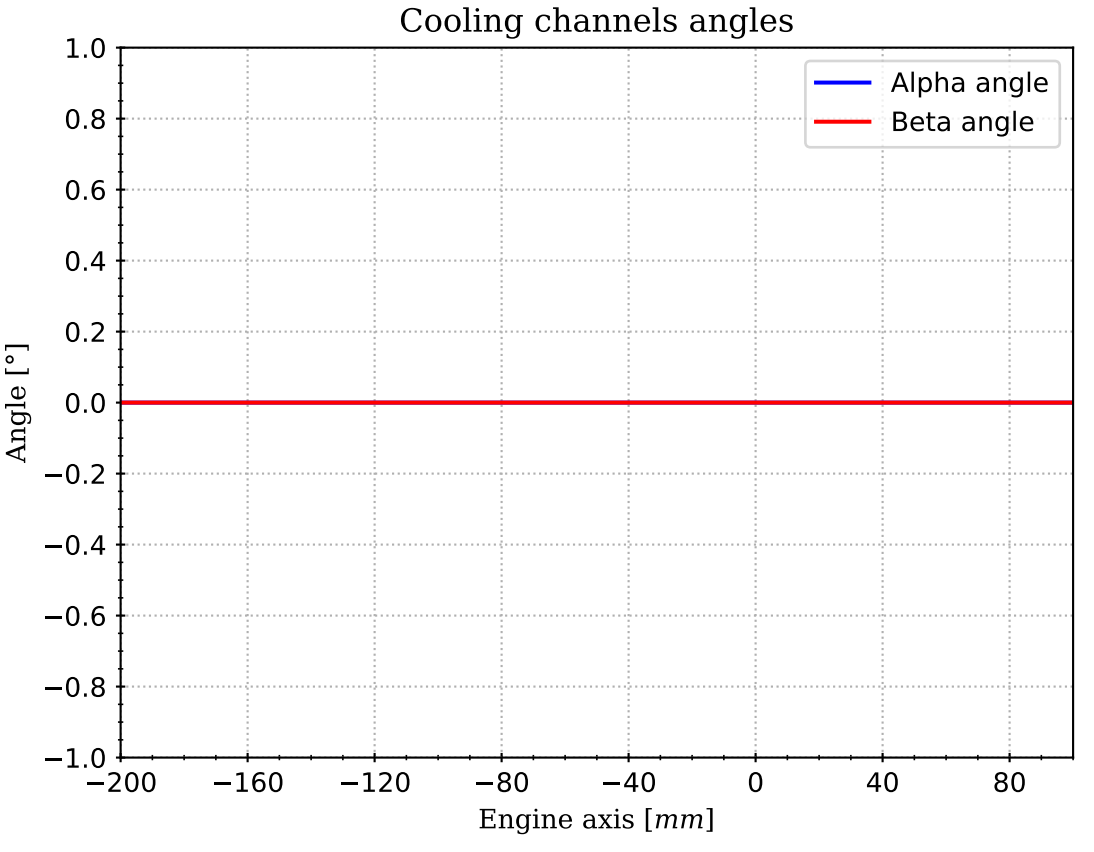

Fin dimensions
Length [mm]
2.7
2.4
2.1
1.8
1.5
1.2
0.9
0.6
0.3
0.0
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Initial fin thickness
Effective fin thickness

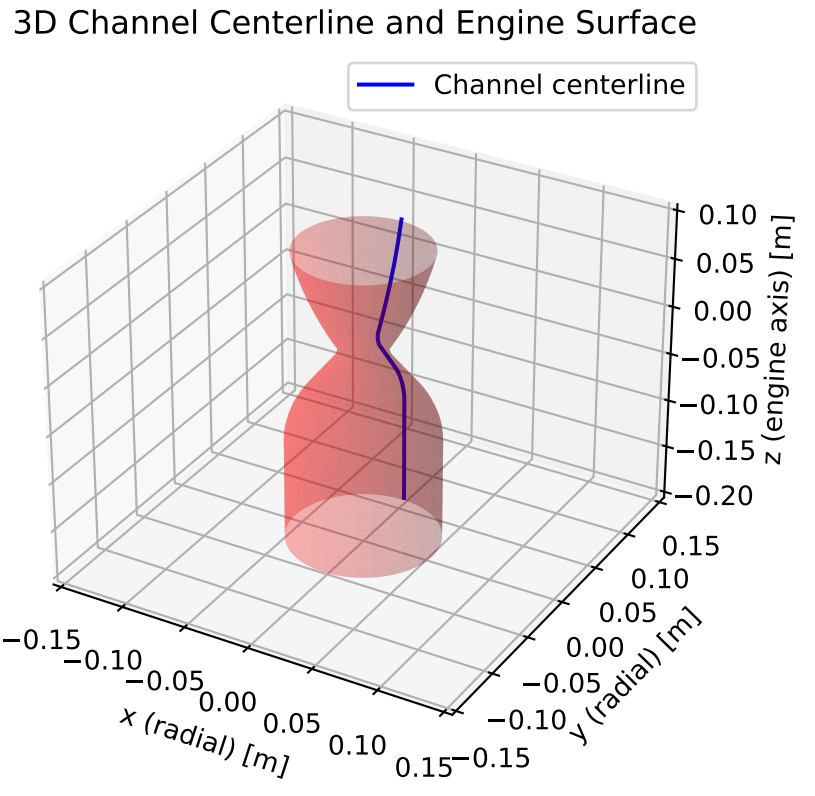

Channel aspect ratio
Aspect Ratio [-]
1.20
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

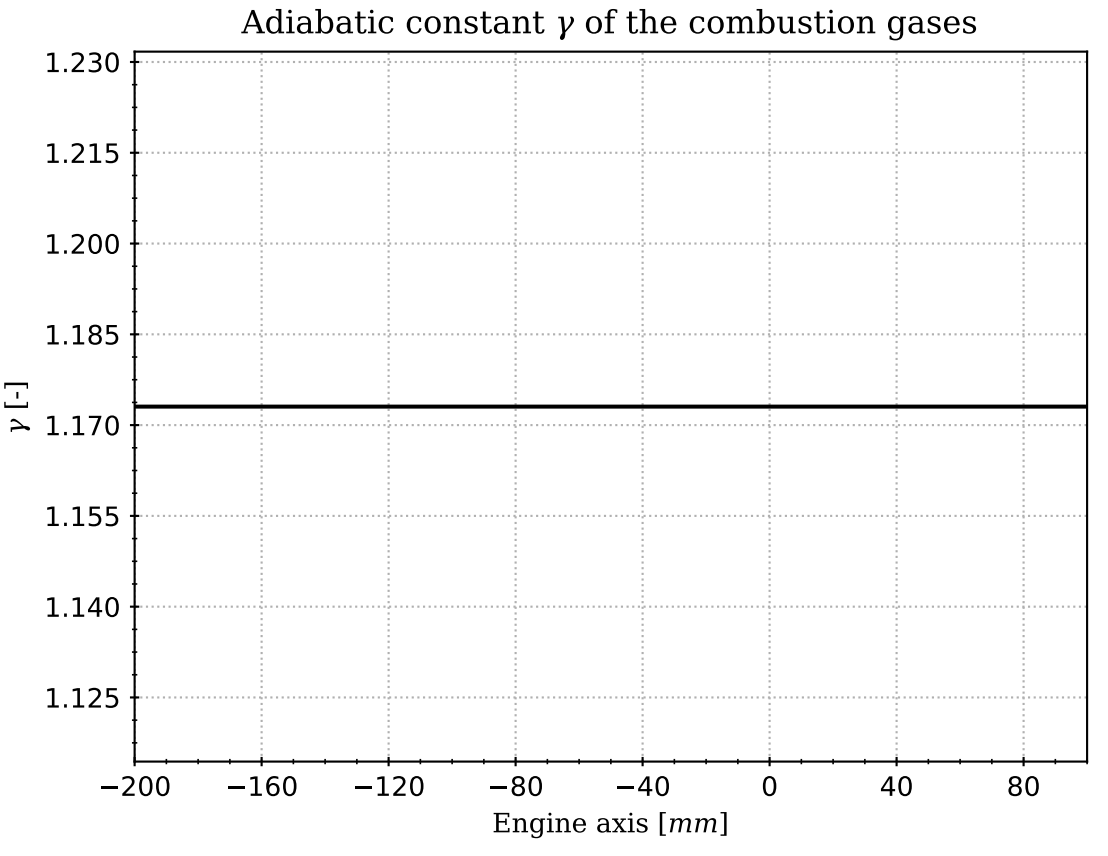

Cooling channels dimensions
Channel height
Channel width
Length [mm]
Engine axis [mm]
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
−200
−160
−120
−80
−40
0
40
80

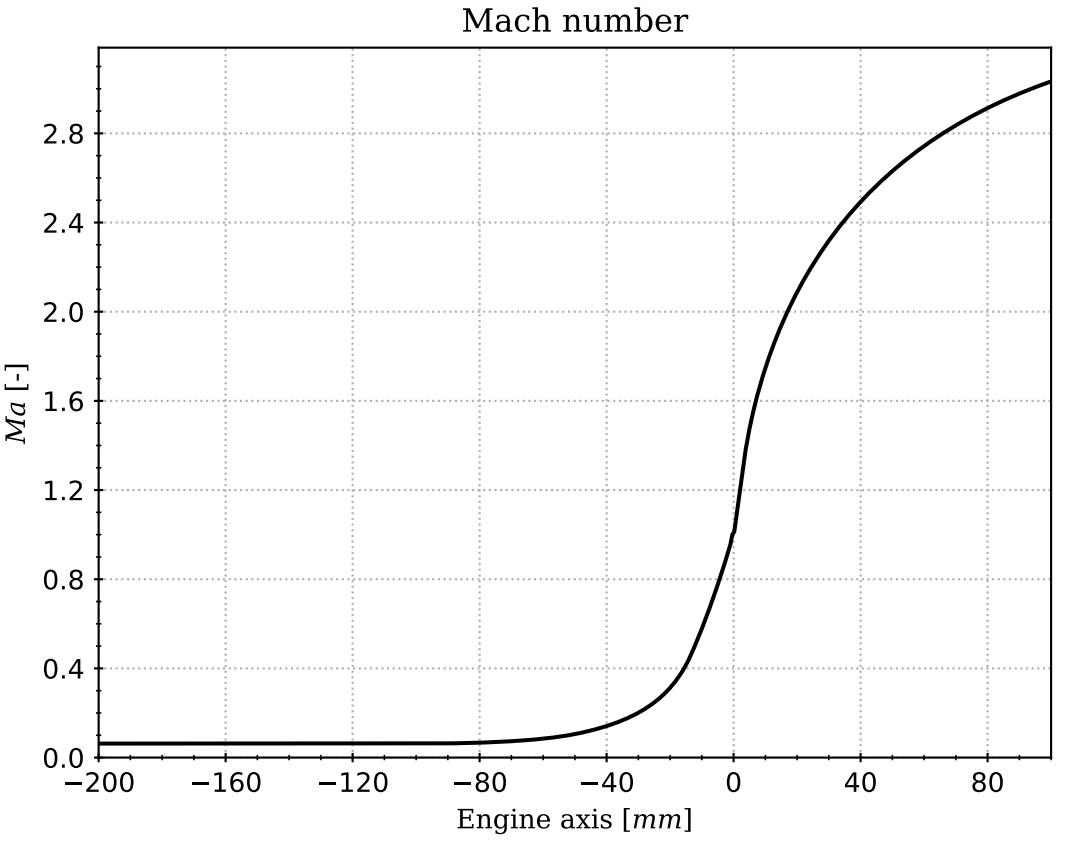

Channel hydraulic diameter
2.7
2.4
2.1
1.8
1.5
1.2
0.9
0.6
0.3
0.0
D_{hy} [mm]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

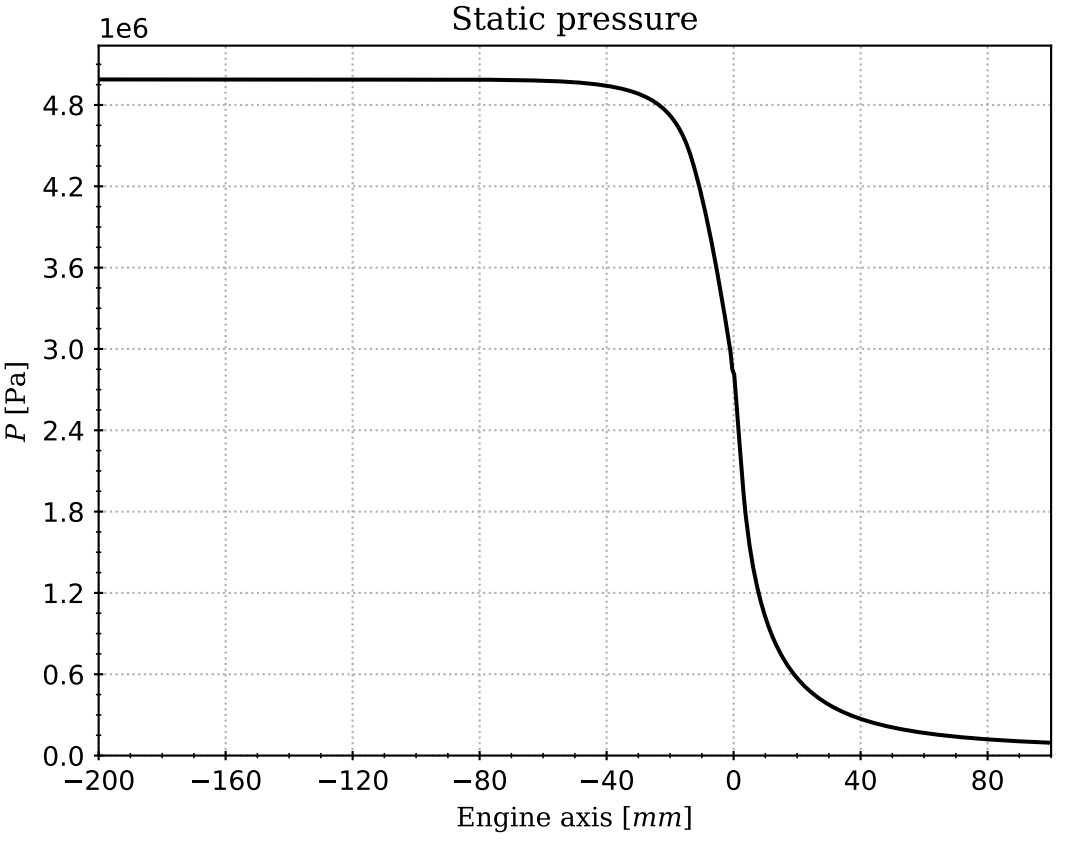

Channel cross-section area
Initial channel cross-section
Effective channel cross-section
Area [mm^2]
Engine axis [mm]
0.0
0.8
1.6
2.4
3.2
4.0
4.8
5.6
6.4
7.2
−200
−160
−120
−80
−40
0
40
80

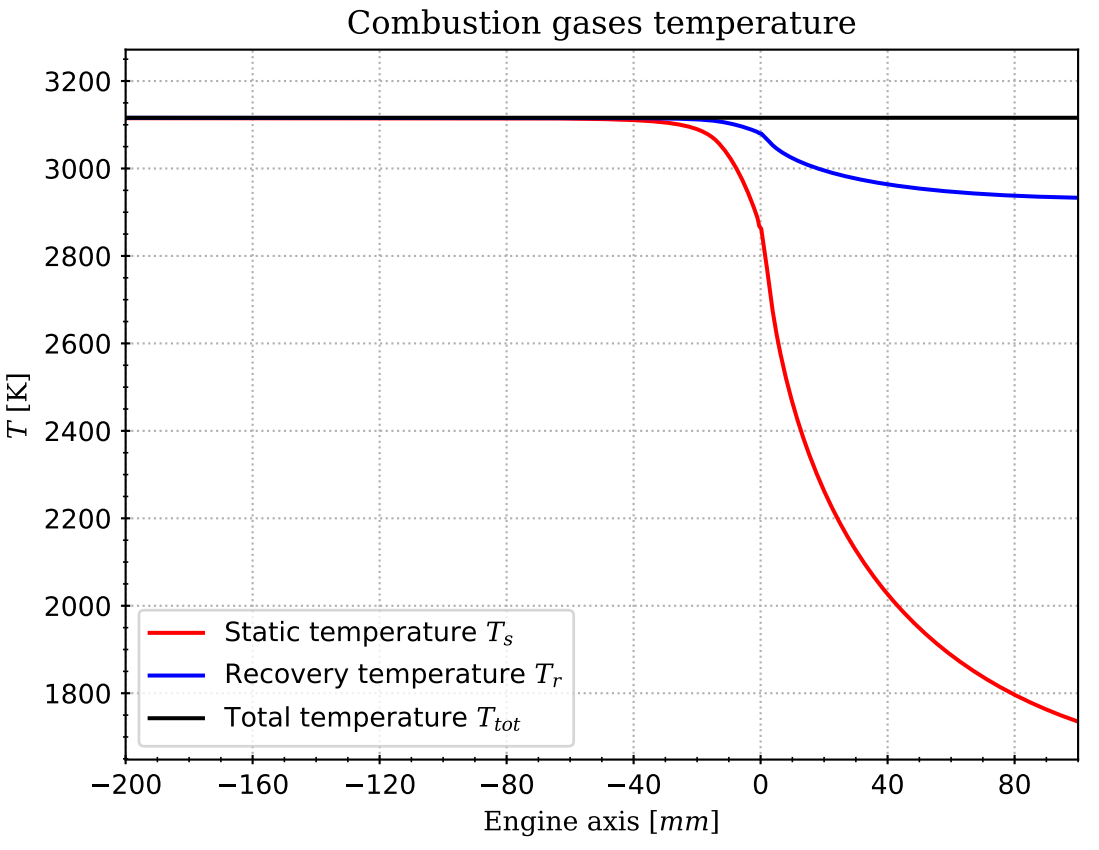

Cooling channels angles
Alpha angle
Beta angle
Angle [°]
Engine axis [mm]
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−200
−160
−120
−80
−40
0
40
80

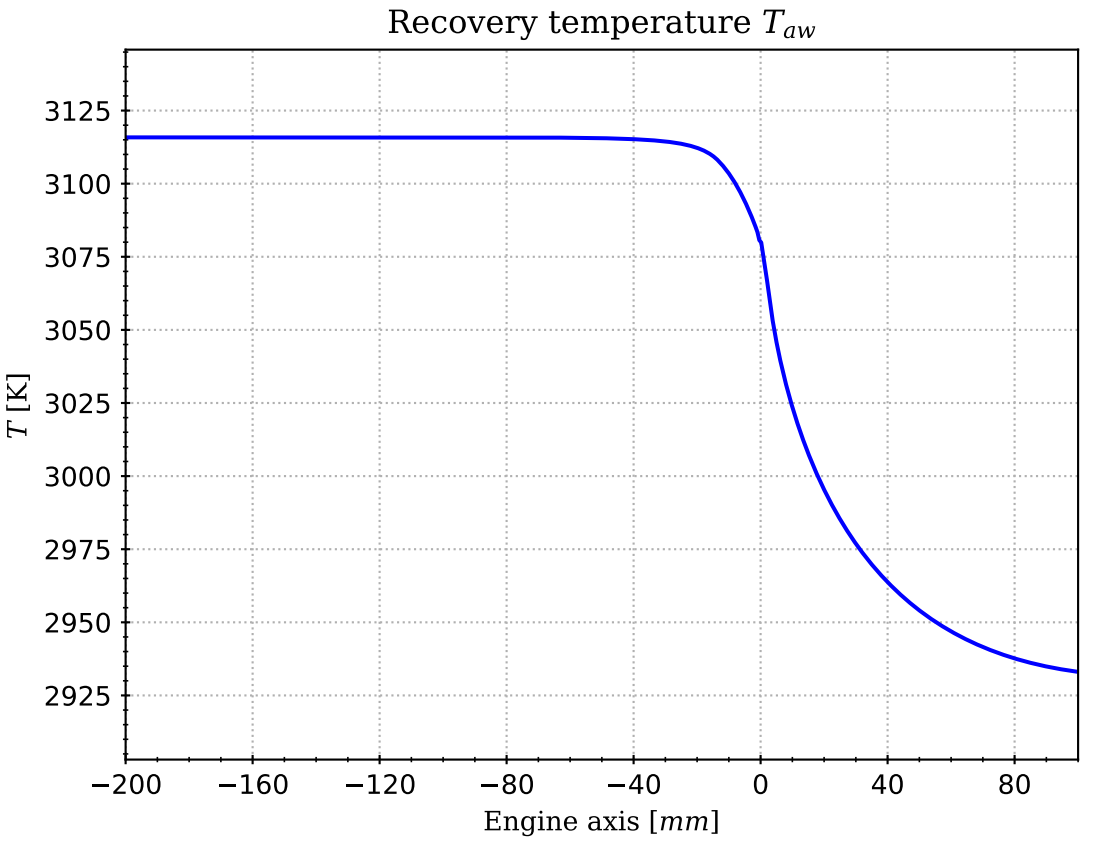

3D Channel Centerline and Engine Surface
Channel centerline
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
z (engine axis) [m]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
y (radial) [m]
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
x (radial) [m]

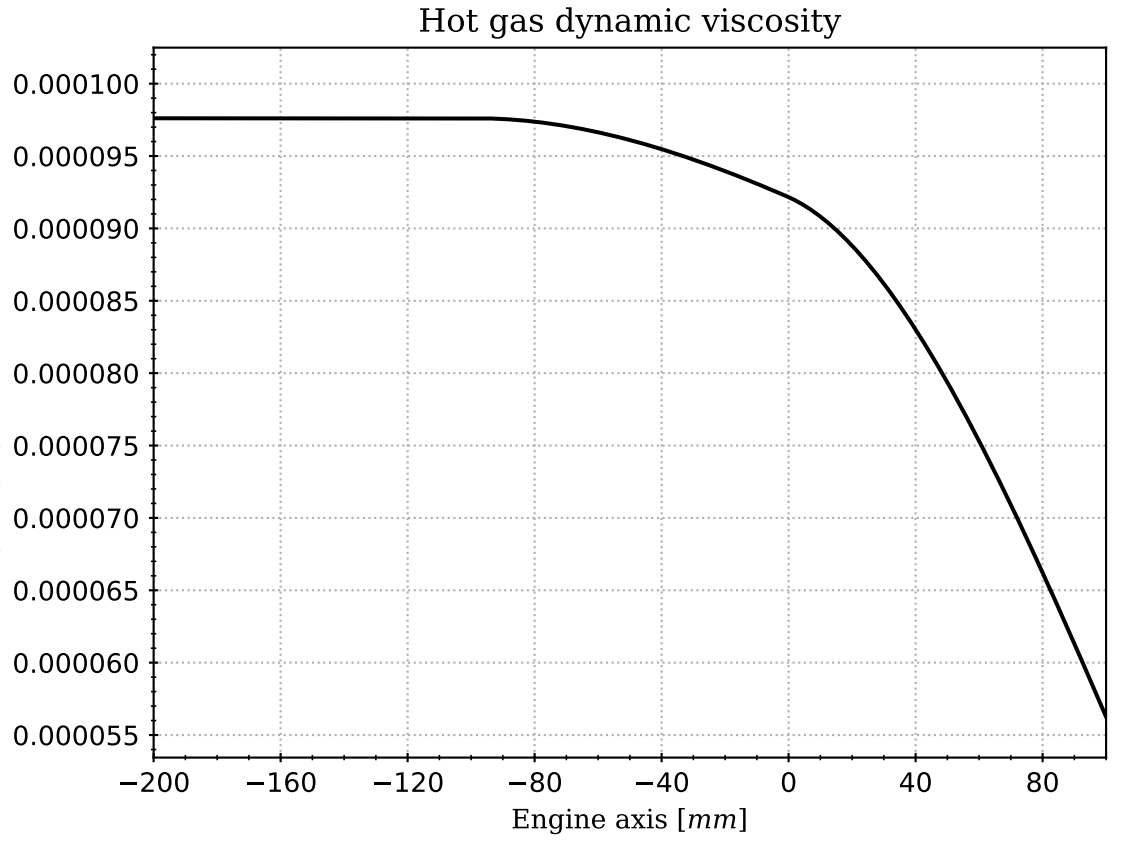

Adiabatic constant γ of the combustion gases
1.230
1.215
1.200
1.185
1.170
1.155
1.140
1.125
γ [-]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

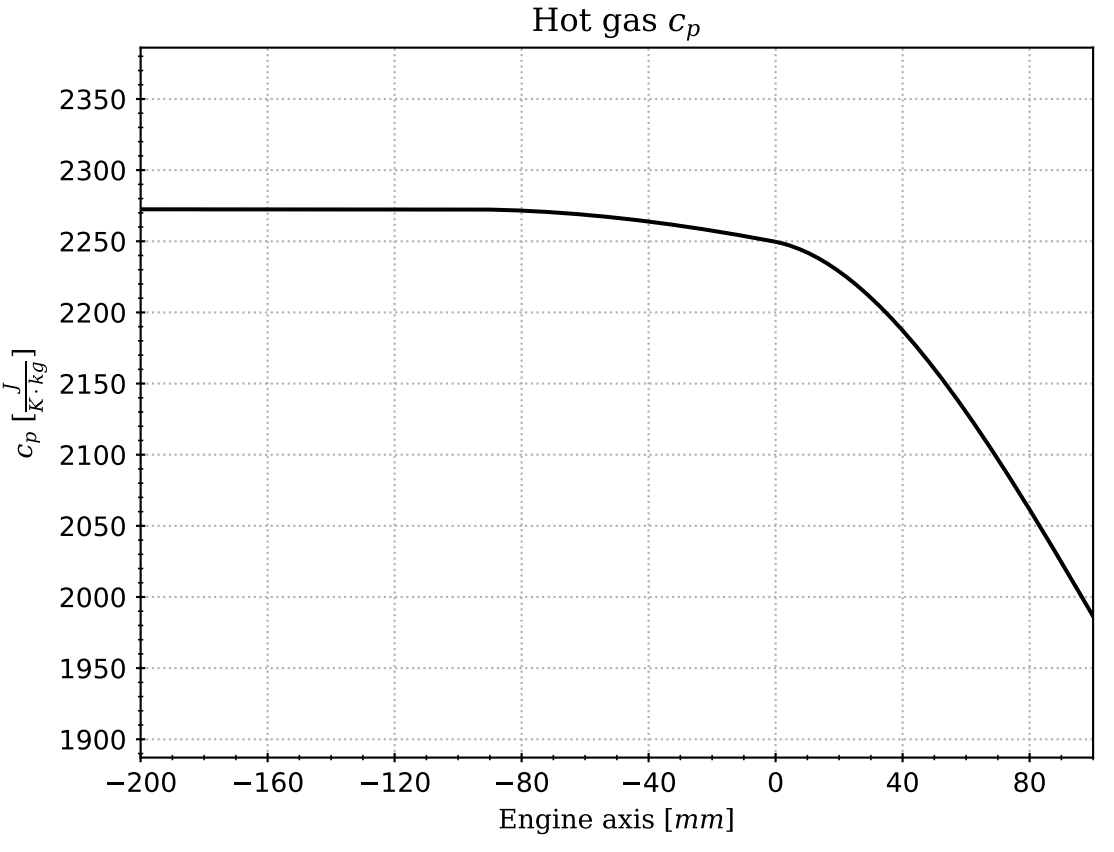

Mach number
Ma [-]
0.0
0.4
0.8
1.2
1.6
2.0
2.4
2.8
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

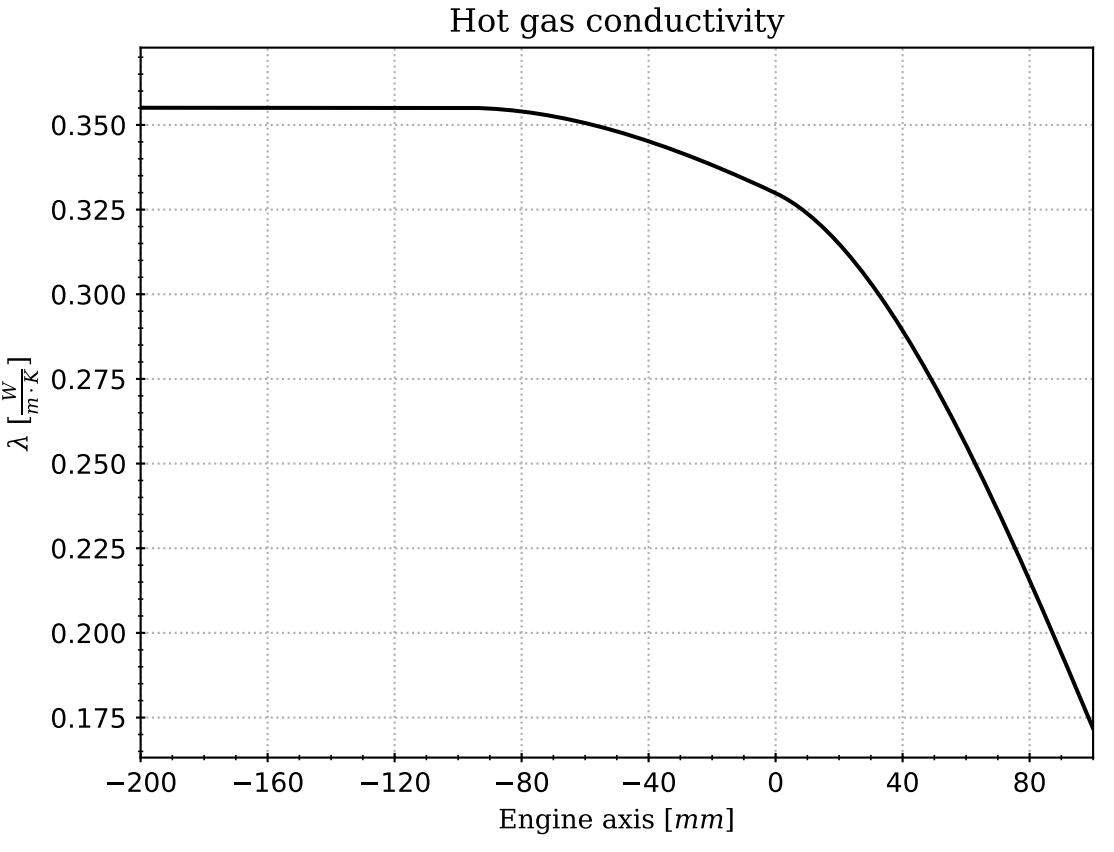

Static pressure
1e6
4.8
4.2
3.6
3.0
2.4
1.8
1.2
0.6
0.0
P [Pa]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

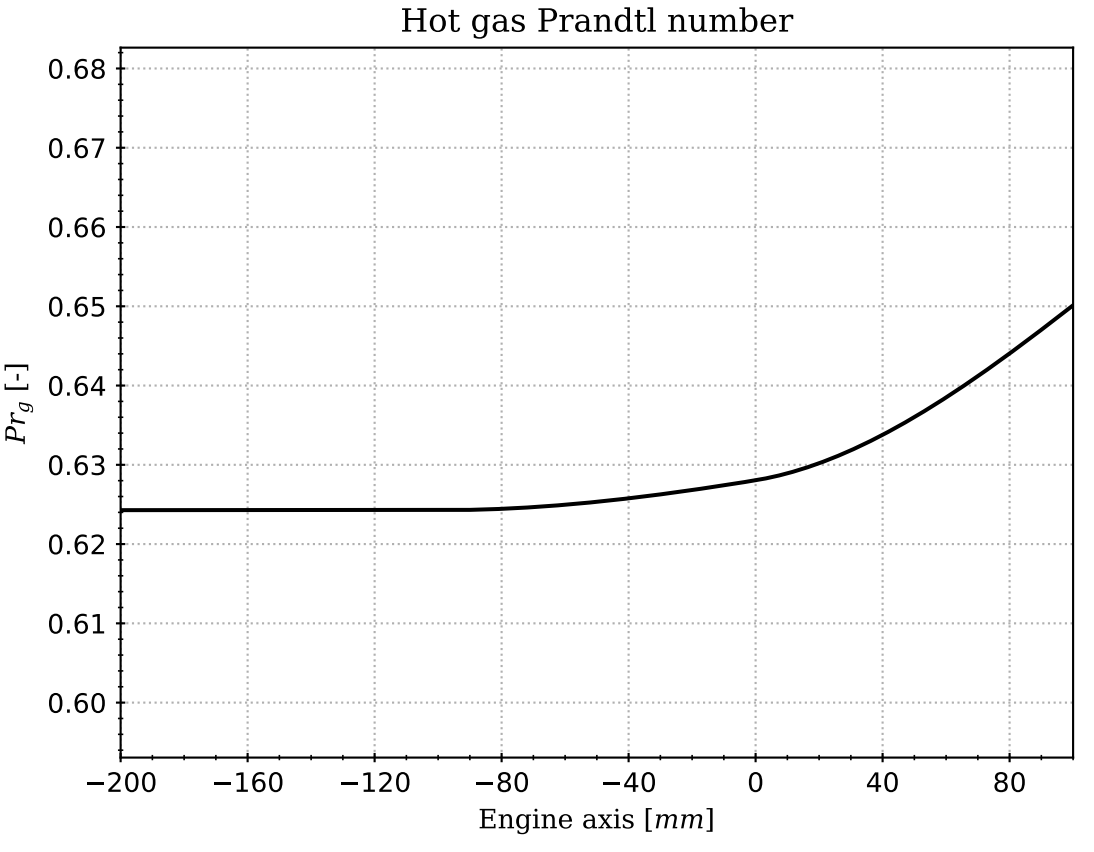

Combustion gases temperature
3200
3000
2800
2600
2400
2200
2000
1800
T [K]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Static temperature T_s
Recovery temperature T_r
Total temperature T_{tot}

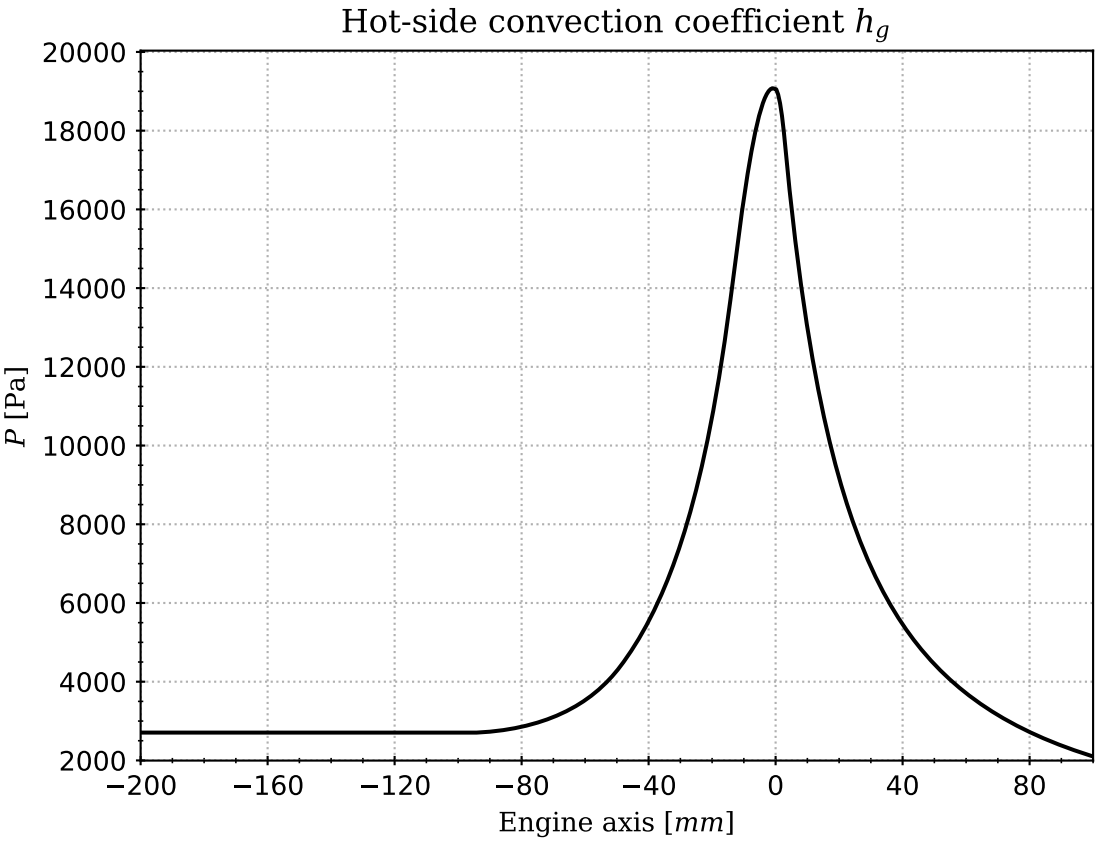

Recovery temperature T_{aw}
3125
3100
3075
3050
3025
3000
2975
2950
2925
T [K]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

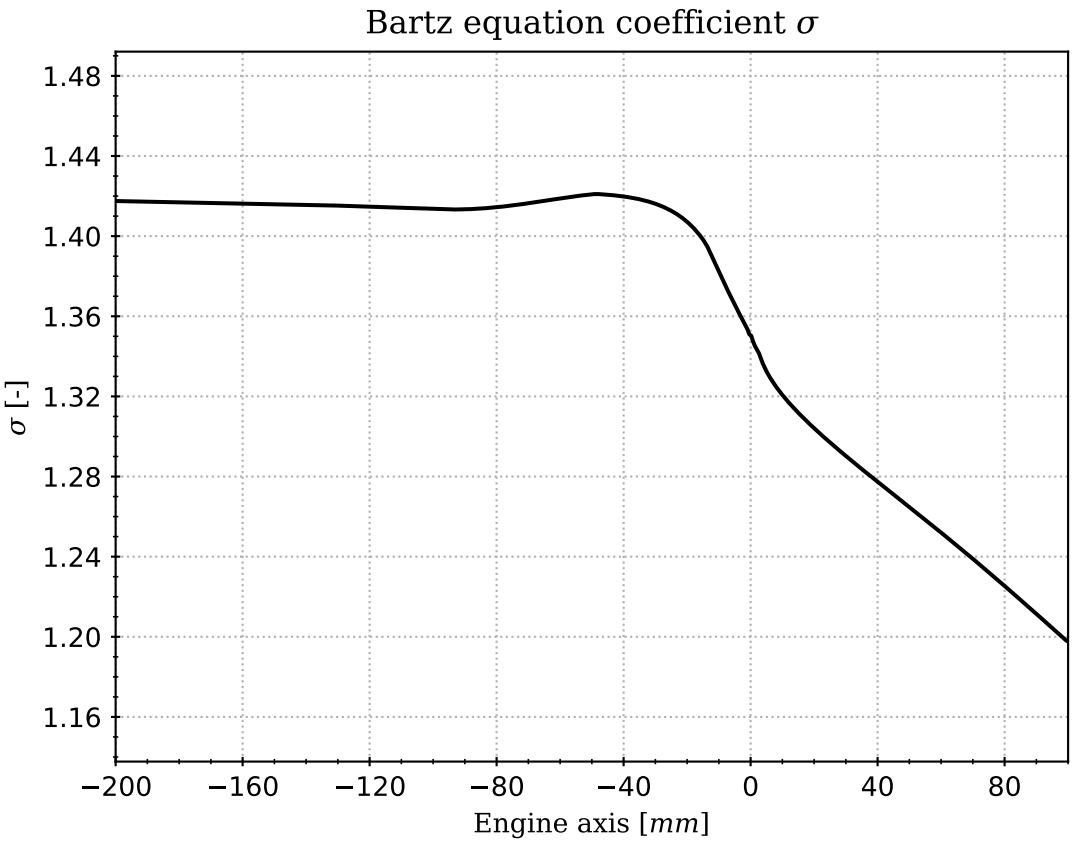

Hot gas dynamic viscosity
0.000100
0.000095
0.000090
0.000085
0.000080
0.000075
0.000070
0.000065
0.000060
0.000055
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

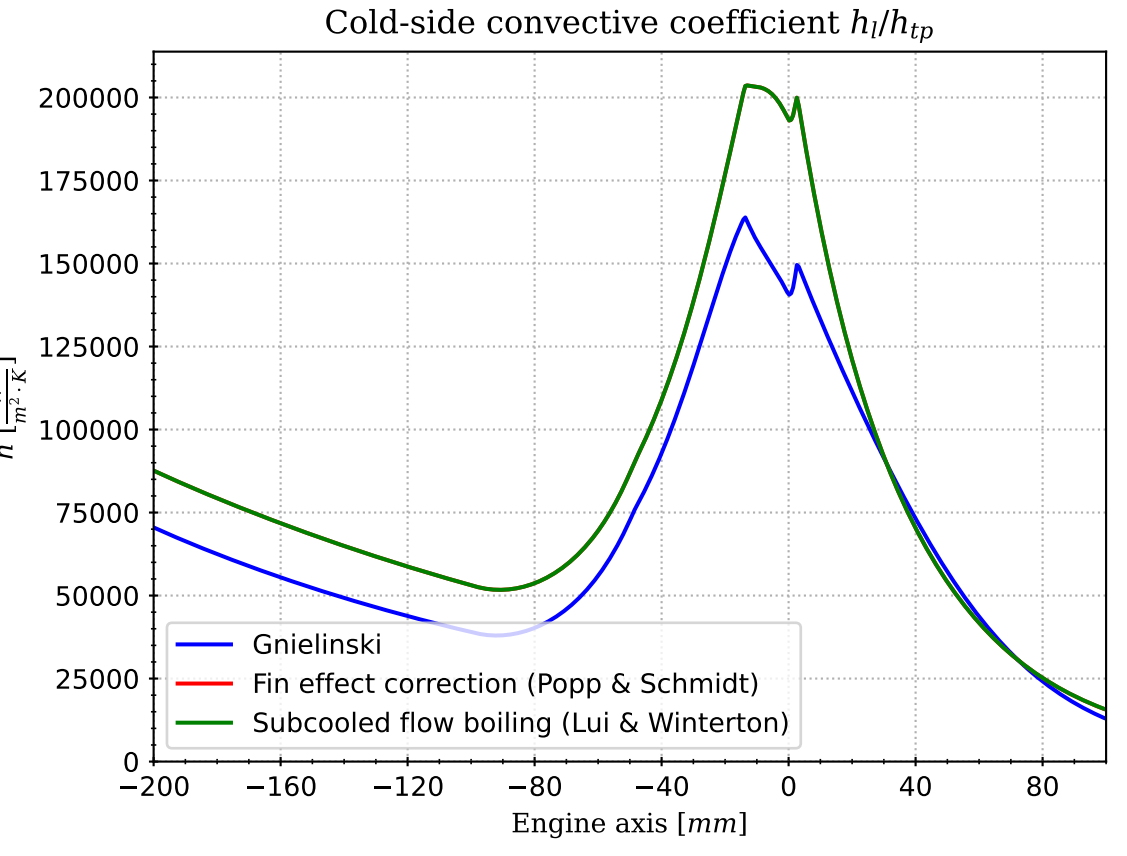

Hot gas c_p
2350
2300
2250
2200
2150
2100
2050
2000
1950
1900
c_p $[\frac{J}{K \cdot kg}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$

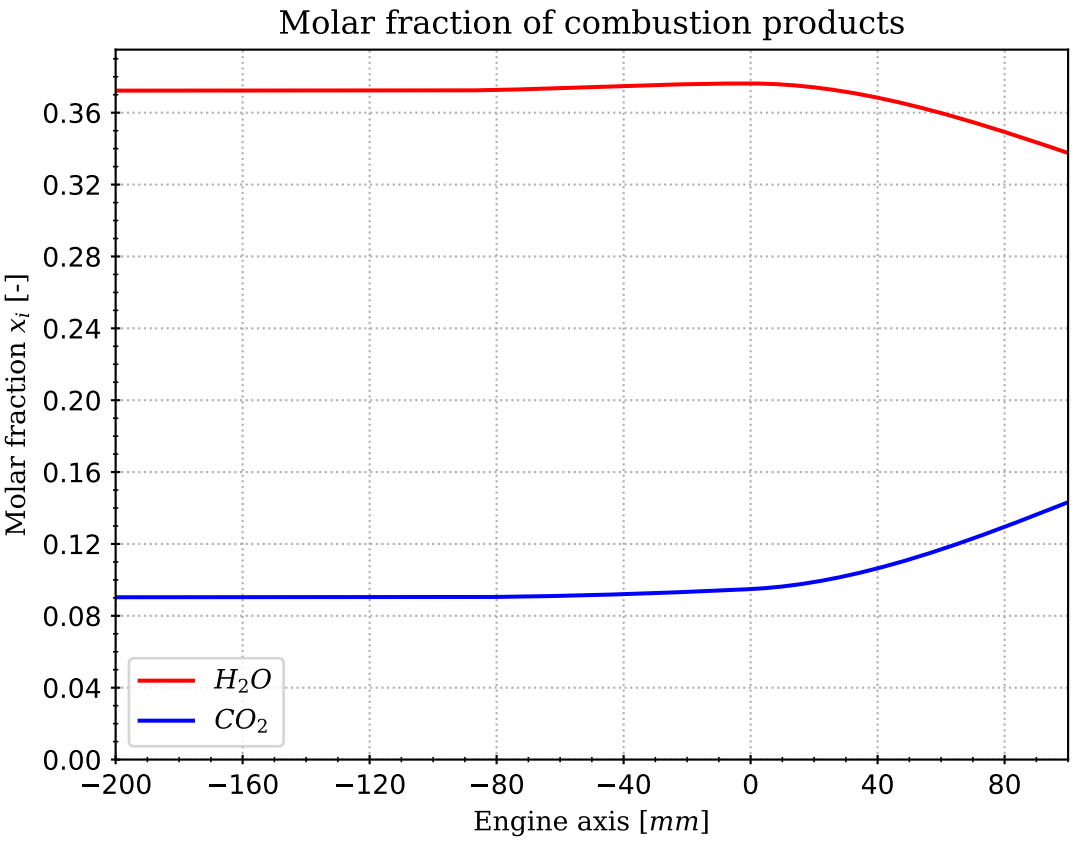

Hot gas conductivity
0.350
0.325
0.300
0.275
0.250
0.225
0.200
0.175
λ $[\frac{W}{m \cdot K}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$

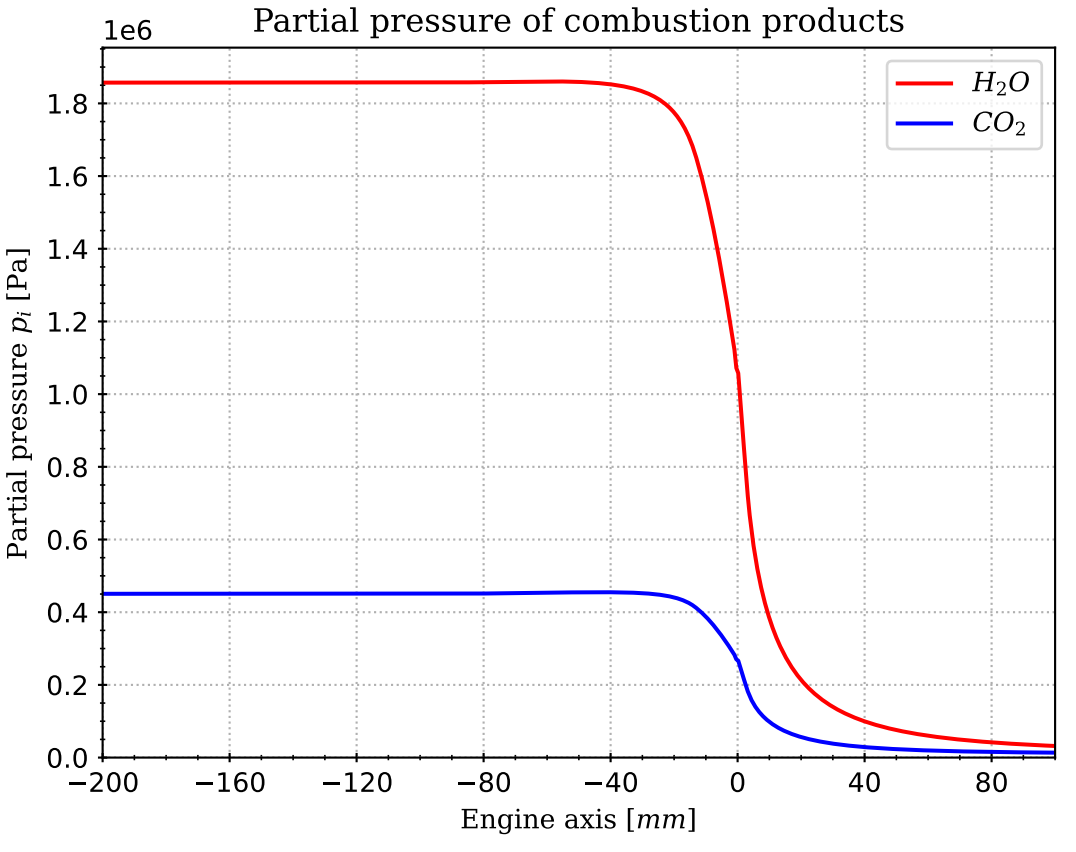

Hot gas Prandtl number
0.68
0.67
0.66
0.65
0.64
0.63
0.62
0.61
0.60
Pr_g [-]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

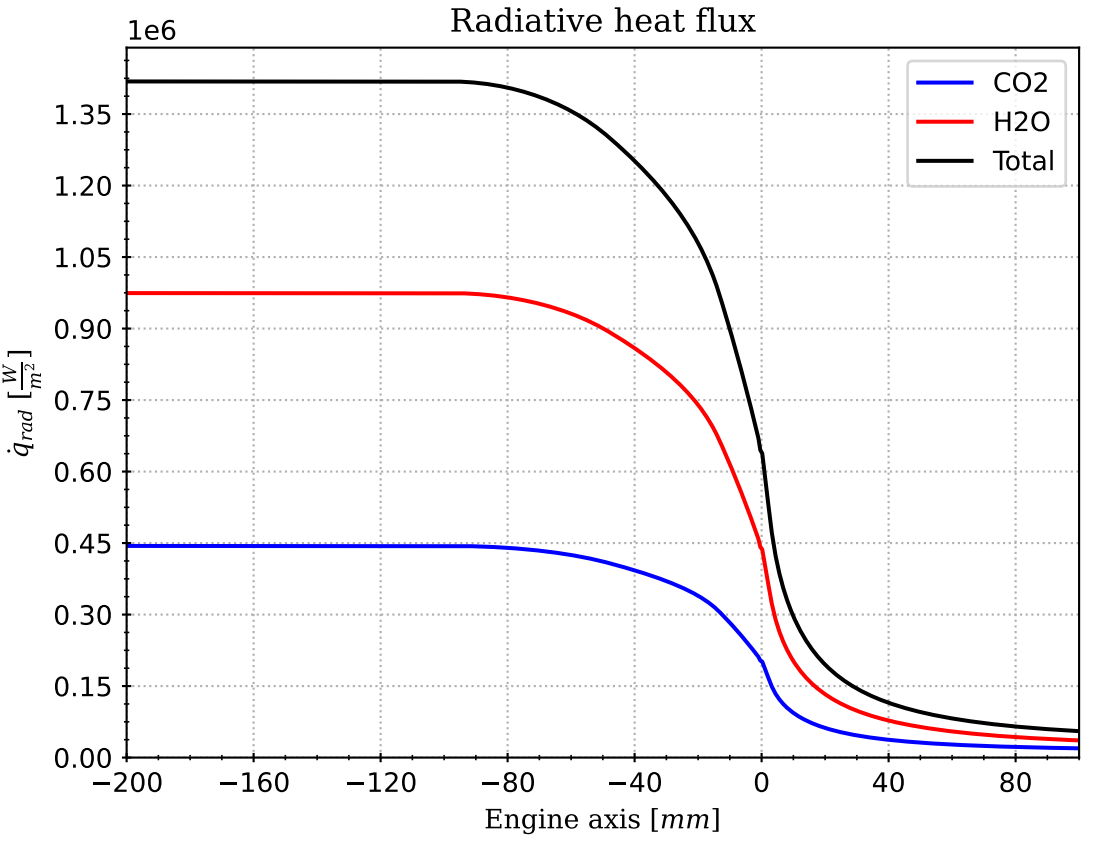

Hot-side convection coefficient h_g
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
P [Pa]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

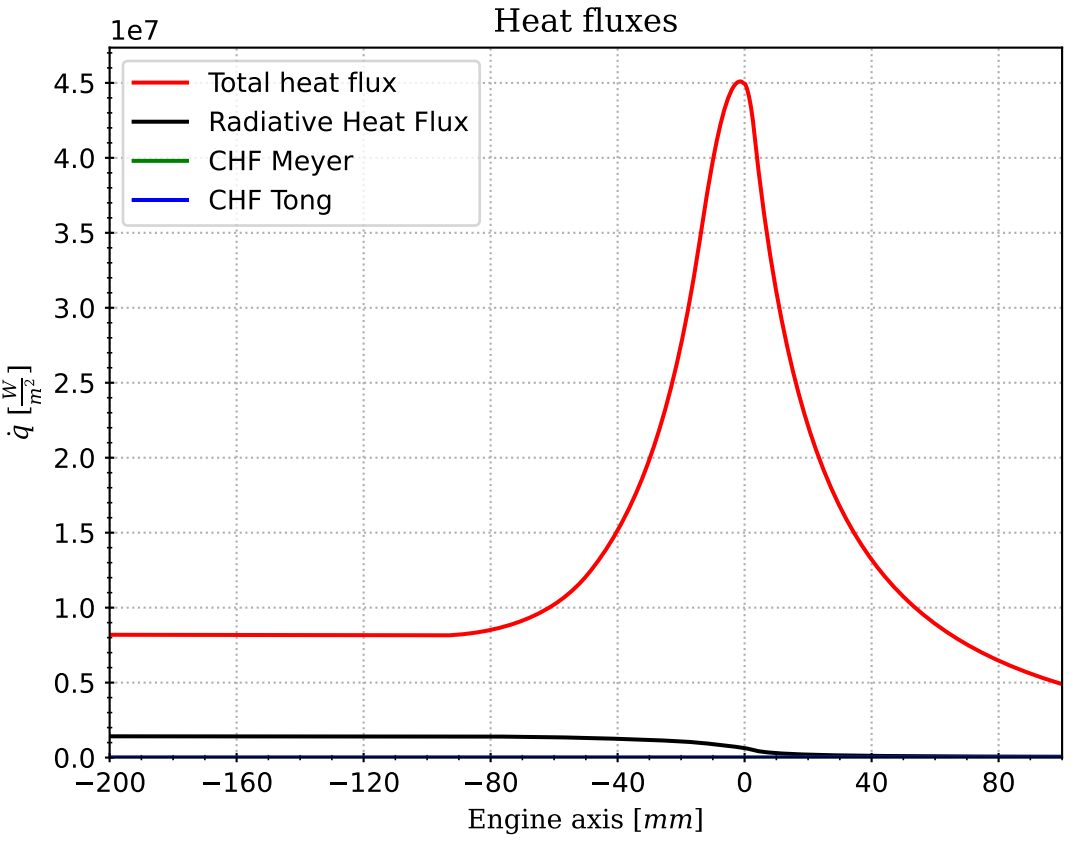

Bartz equation coefficient σ
1.48
1.44
1.40
1.36
1.32
1.28
1.24
1.20
1.16
σ [-]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

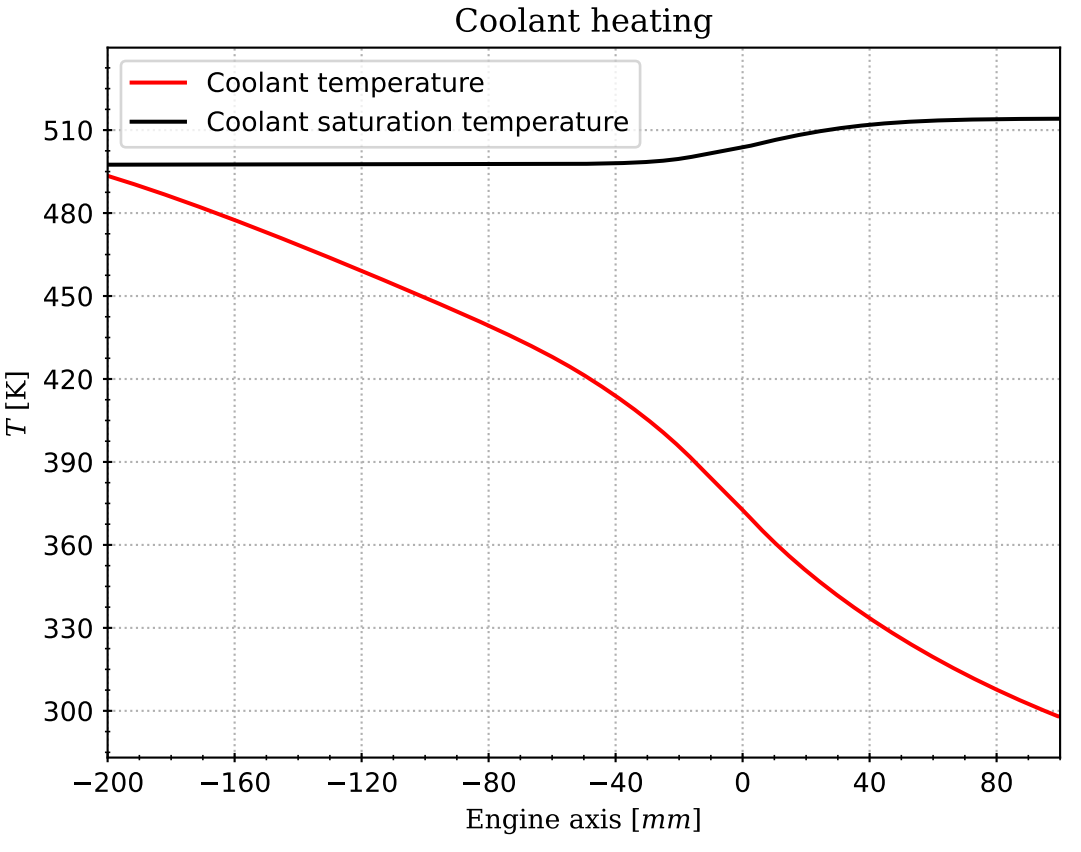

Cold-side convective coefficient h_l/h_{tp}
200000
175000
150000
125000
100000
75000
50000
25000
0
$h\ [\frac{W}{m^2 \cdot K}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]
Gnielinski
Fin effect correction (Popp & Schmidt)
Subcooled flow boiling (Lui & Winterton)

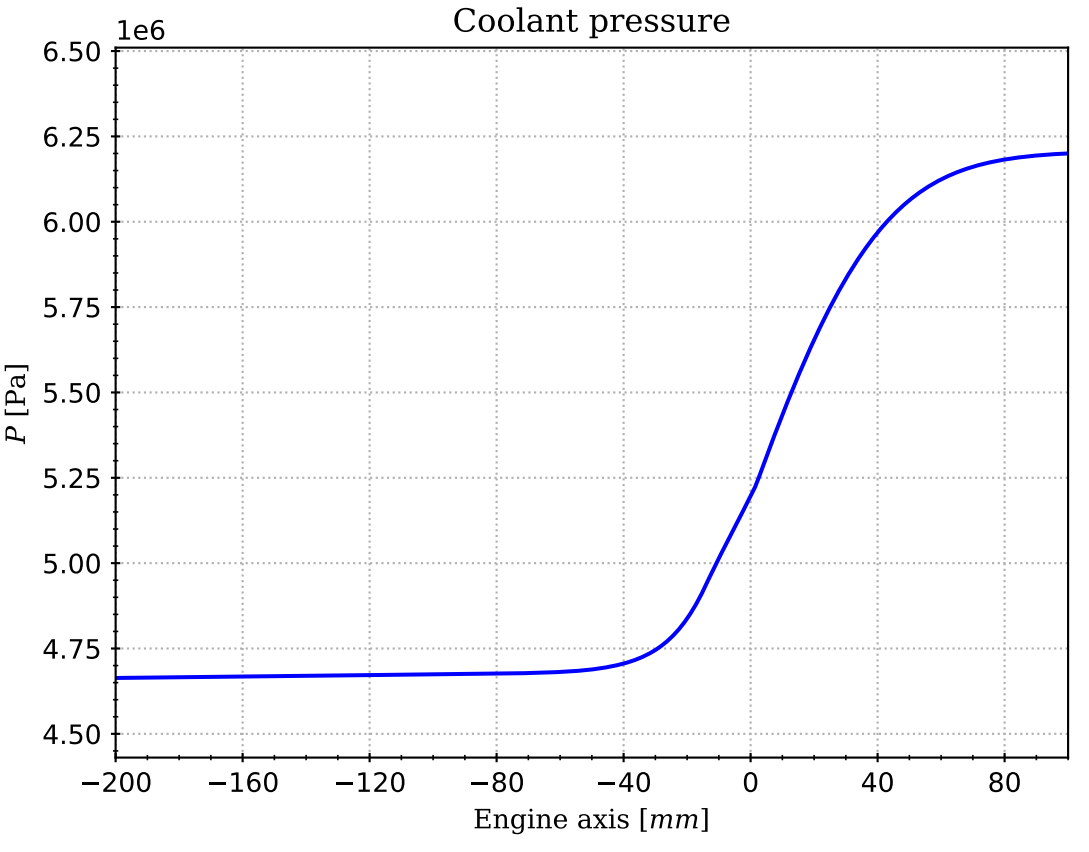

Molar fraction of combustion products
Molar fraction x_i [-]
0.36
0.32
0.28
0.24
0.20
0.16
0.12
0.08
0.04
0.00
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]
H_2O
CO_2

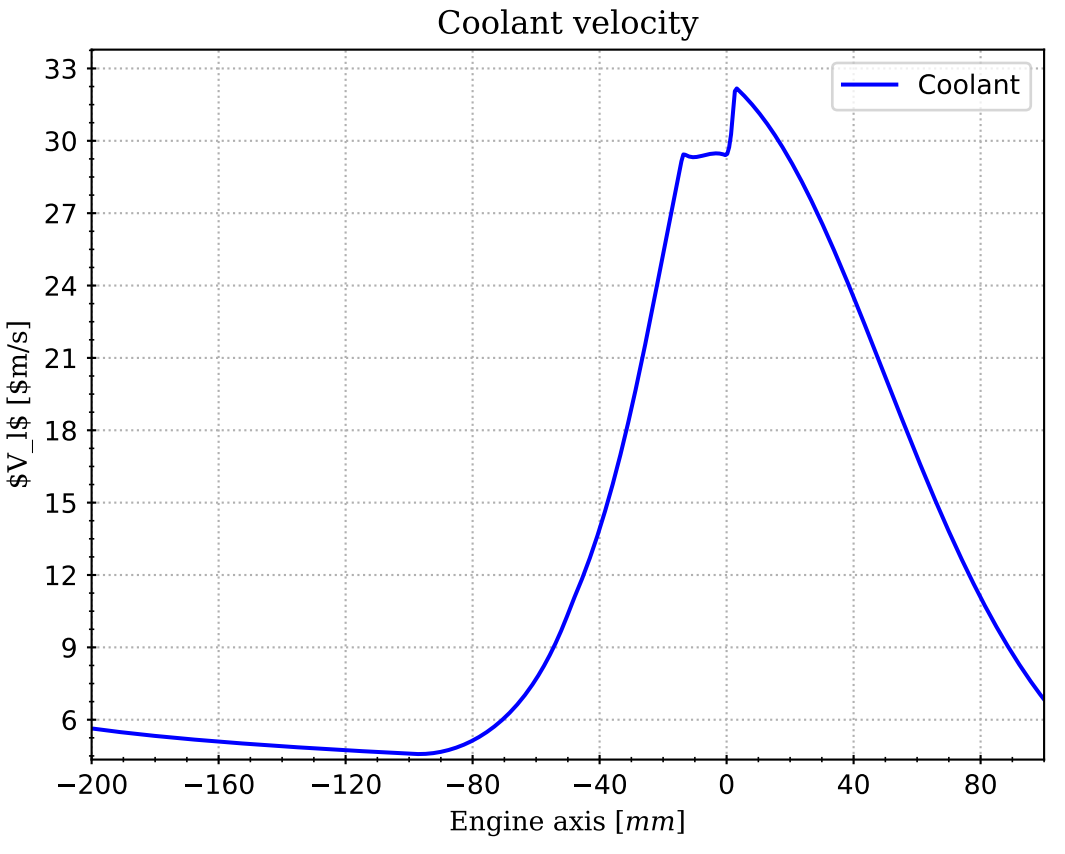

Partial pressure of combustion products
1e6
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Partial pressure p_i [Pa]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]
H_2O
CO_2

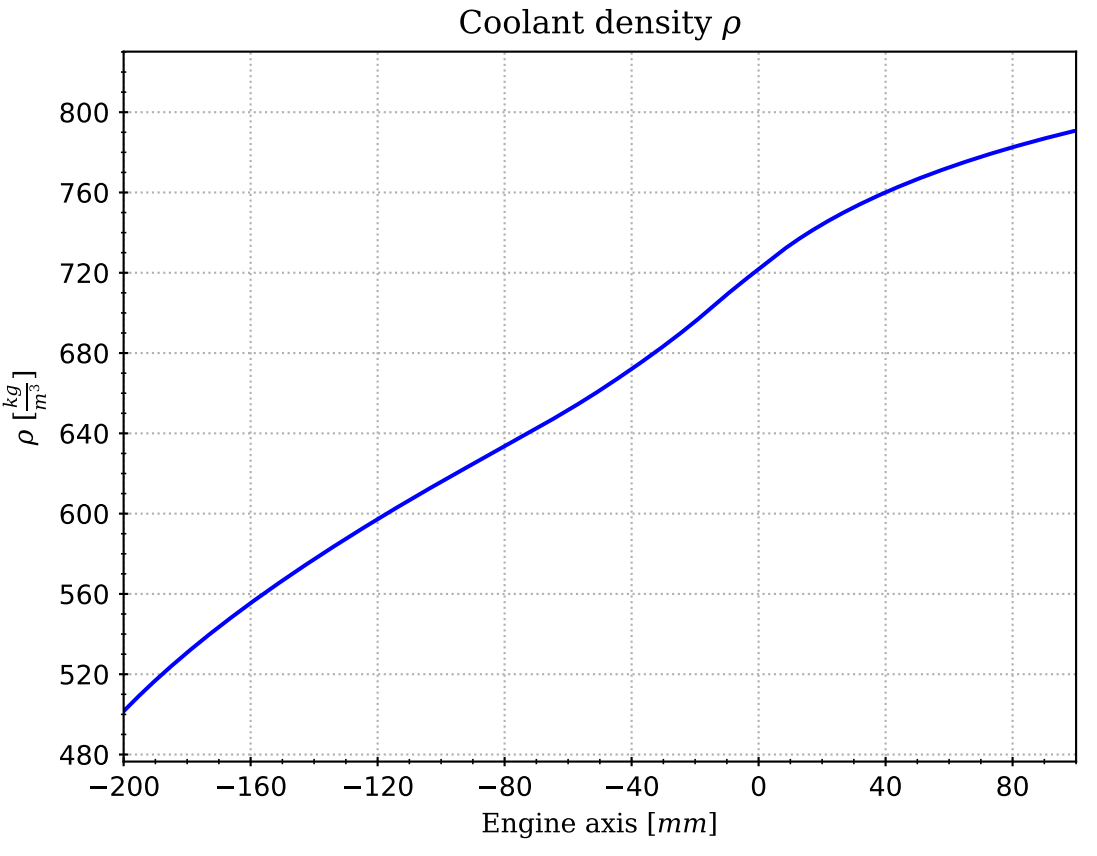

Radiative heat flux
1e6
1.35
1.20
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
$\dot{q}_{rad}$ $[\frac{W}{m^2}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$
CO2
H2O
Total

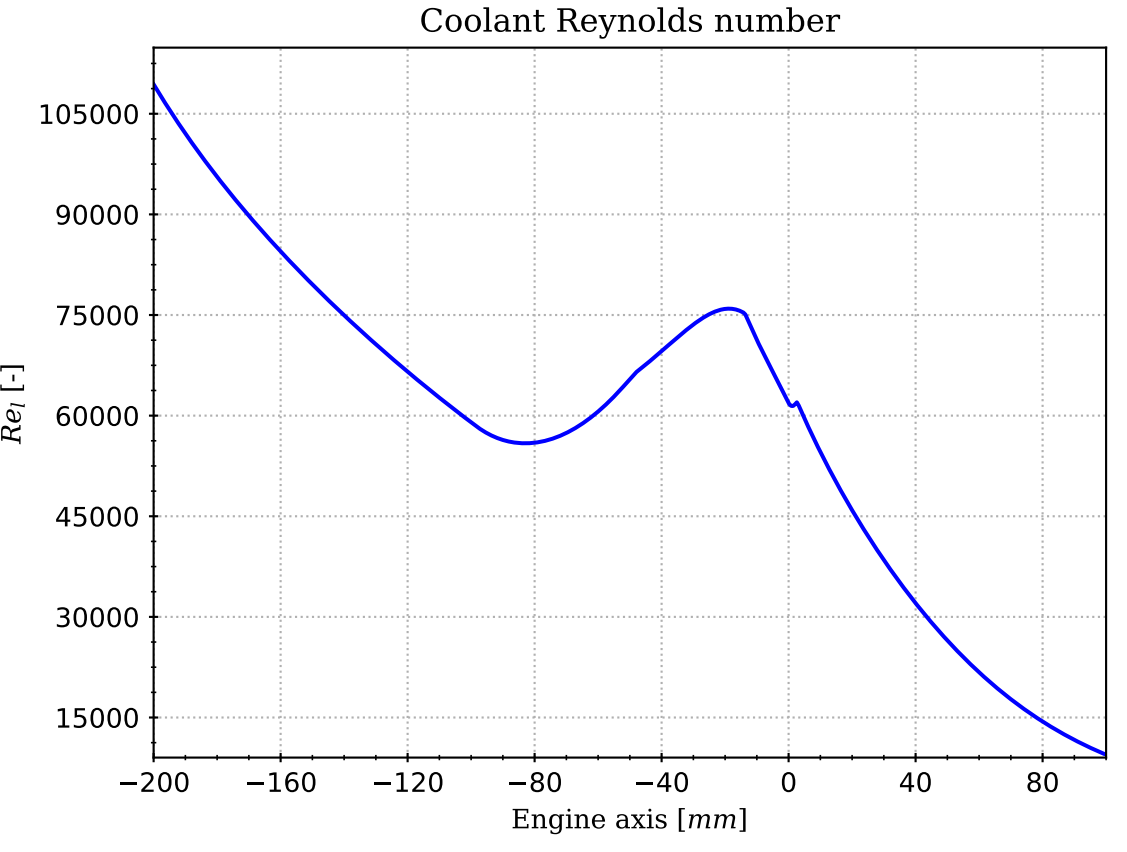

Heat fluxes
1e7
Total heat flux
Radiative Heat Flux
CHF Meyer
CHF Tong
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
$\dot{q}\ [\frac{W}{m^2}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$

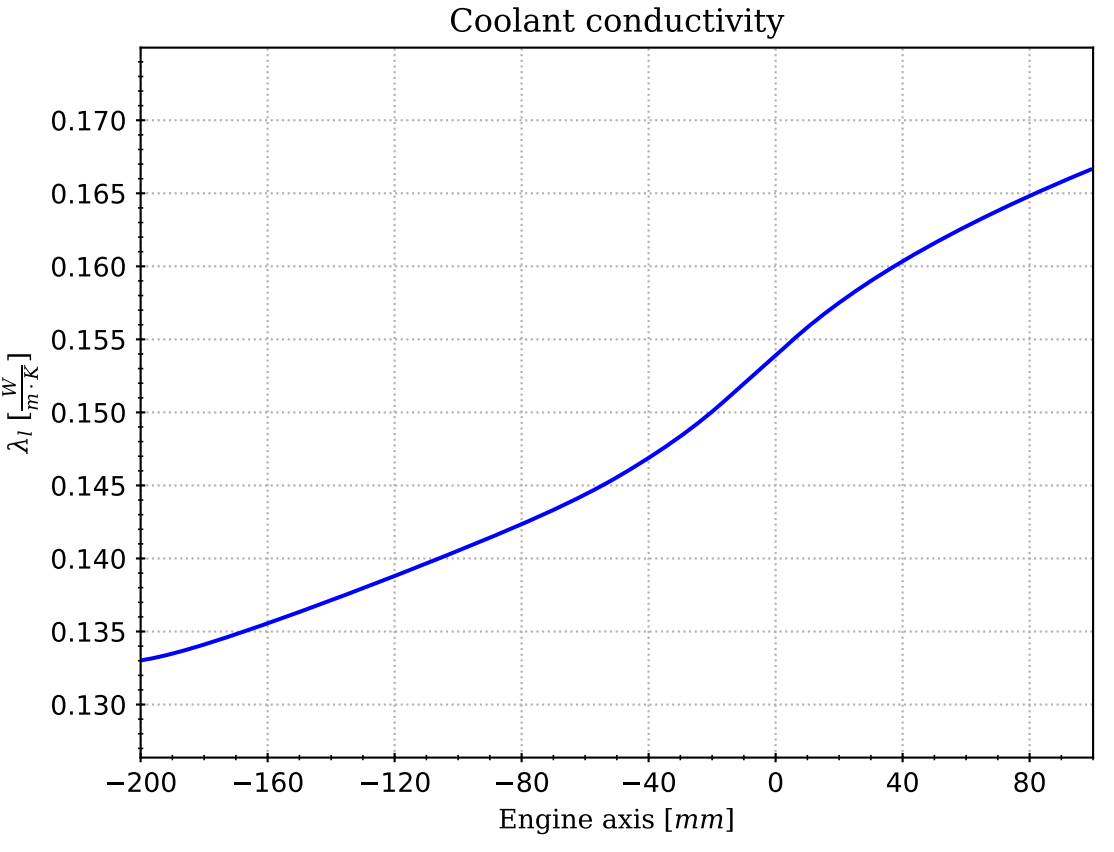

Coolant heating
Coolant temperature
Coolant saturation temperature
510
480
450
420
390
360
330
300
T [K]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

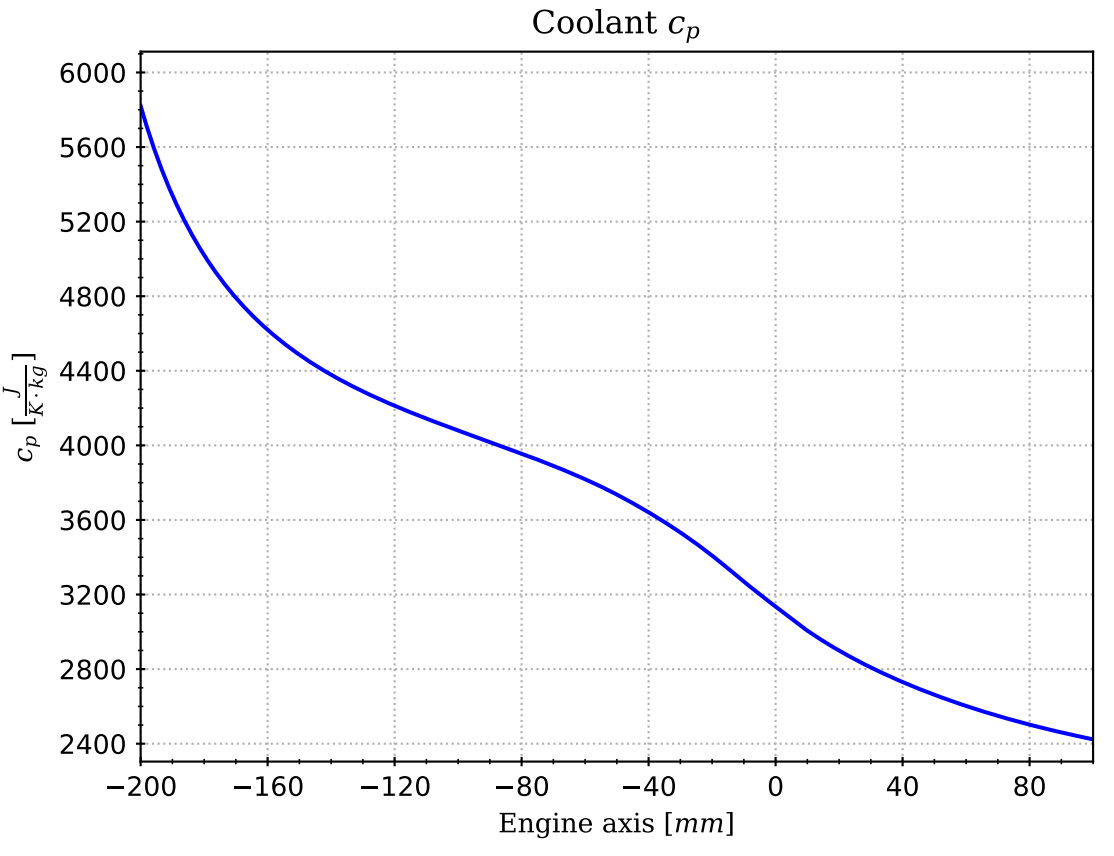

Coolant pressure
1e6
6.50
6.25
6.00
5.75
5.50
5.25
5.00
4.75
4.50
P [Pa]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

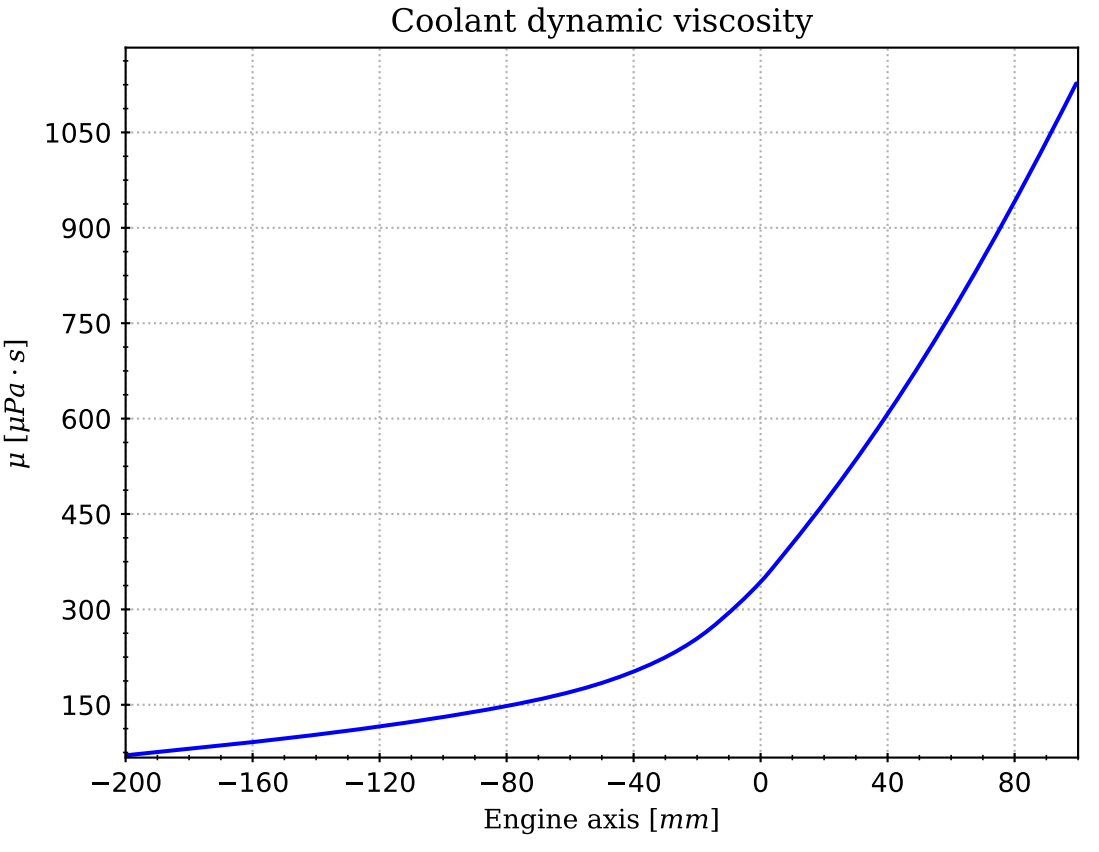

Coolant velocity
Coolant
33
30
27
24
21
18
15
12
9
6
V_l [$m/s]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

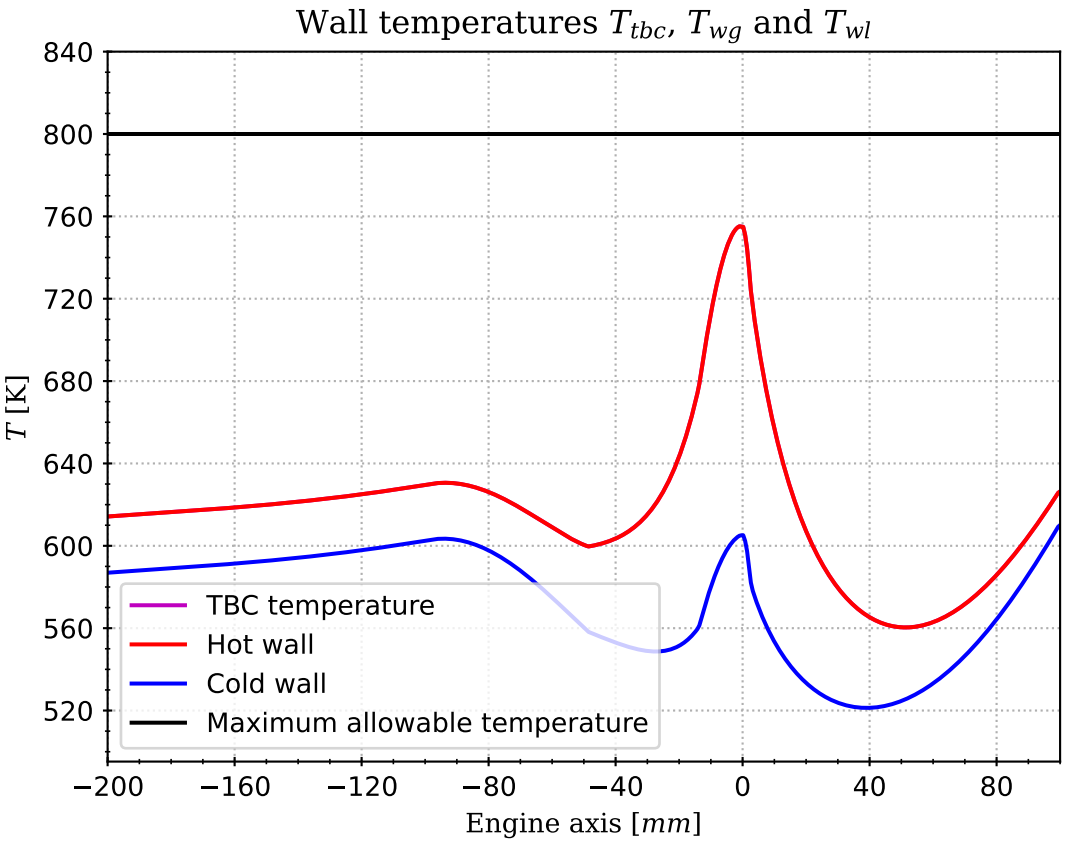

Coolant density ρ
800
760
720
680
640
600
560
520
480
$\rho\ [\frac{kg}{m^3}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$

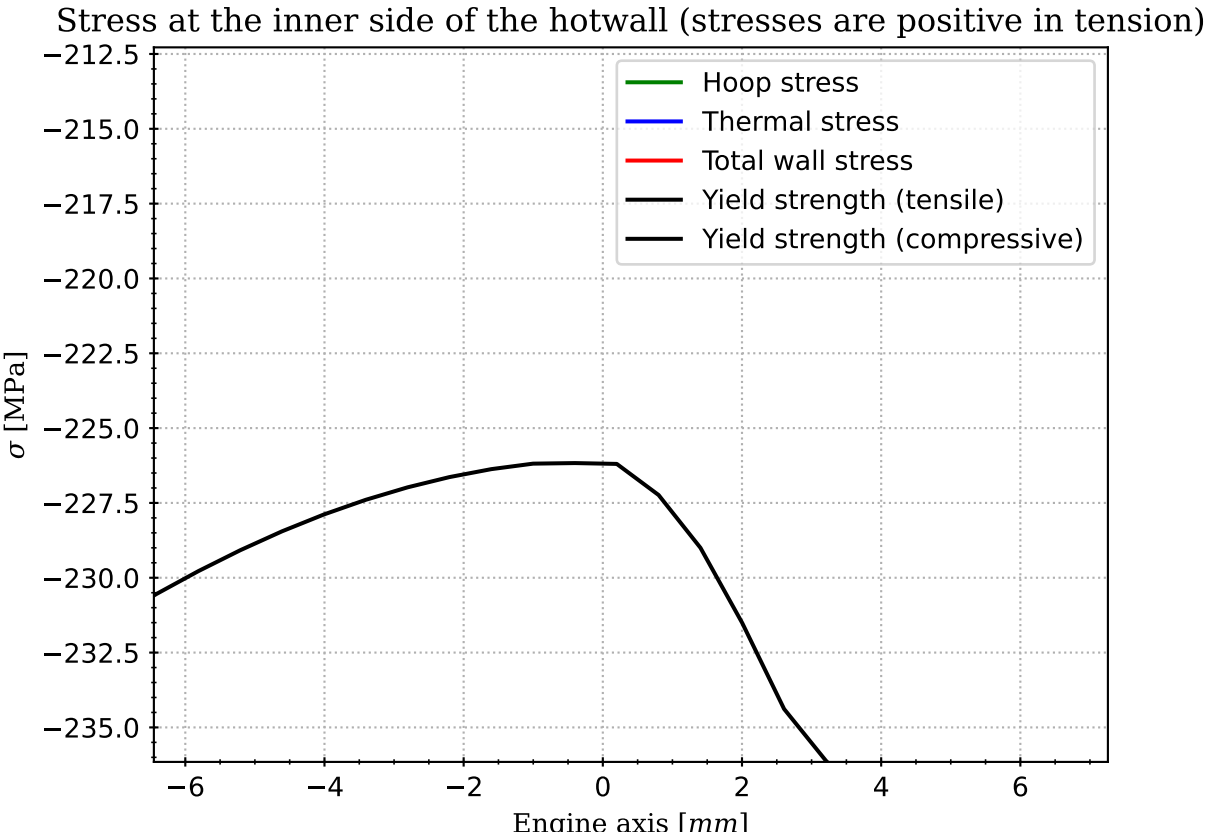

Coolant Reynolds number
105000
90000
75000
60000
45000
30000
15000
Re_l [-]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]

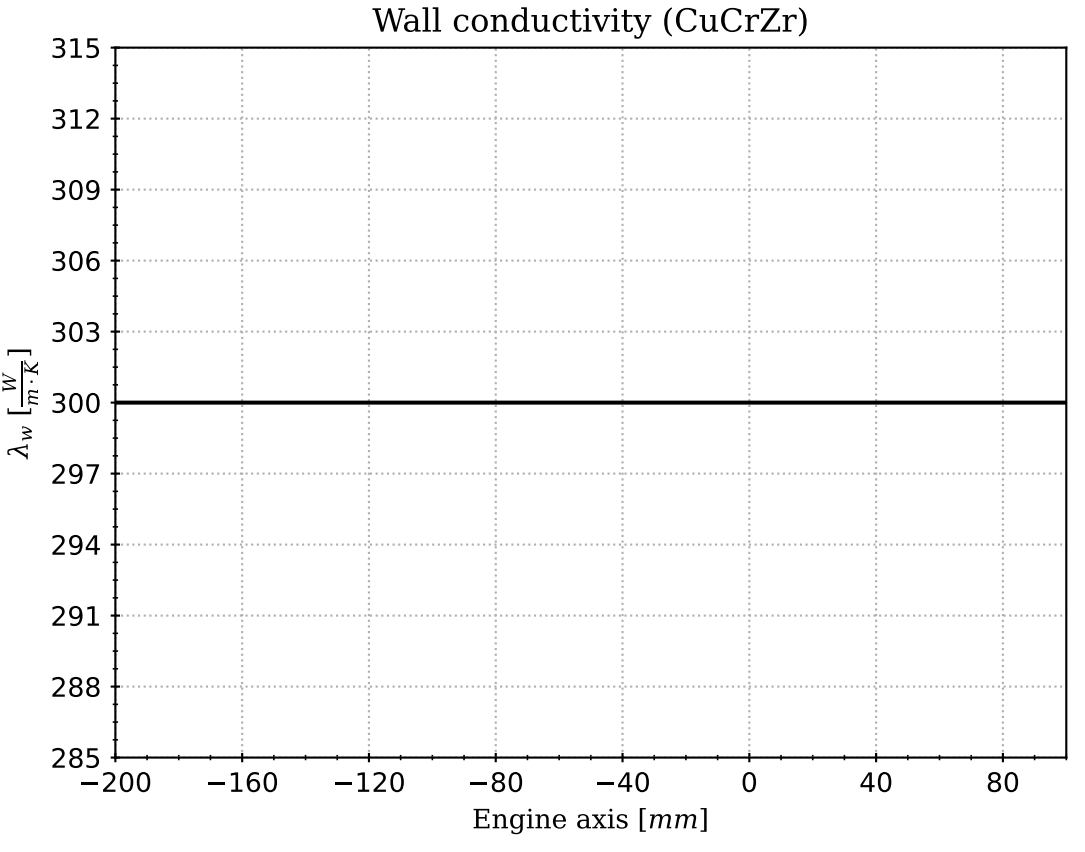

Coolant conductivity
0.170
0.165
0.160
0.155
0.150
0.145
0.140
0.135
0.130
λ_l $[\frac{W}{m \cdot K}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$


Coolant c_p
6000
5600
5200
4800
4400
4000
3600
3200
2800
2400
c_p $[\frac{J}{K \cdot kg}]$
−200
−160
−120
−80
−40
0
40
80
Engine axis $[mm]$


Coolant dynamic viscosity
1050
900
750
600
450
300
150
μ [$\mu Pa \cdot s$]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]


Wall temperatures T_{tbc}, T_{wg} and T_{wl}
840
800
760
720
680
640
600
560
520
T [K]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]
TBC temperature
Hot wall
Cold wall
Maximum allowable temperature


Stress at the inner side of the hotwall (stresses are positive in tension)
Hoop stress
Thermal stress
Total wall stress
Yield strength (tensile)
Yield strength (compressive)
−212.5
−215.0
−217.5
−220.0
−222.5
−225.0
−227.5
−230.0
−232.5
−235.0
σ [MPa]
−6
−4
−2
0
2
4
6
Engine axis [mm]


Wall conductivity (CuCrZr)
315
312
309
306
303
300
297
294
291
288
285
λ_w [$\frac{W}{m \cdot K}$]
−200
−160
−120
−80
−40
0
40
80
Engine axis [mm]